

PRABHU SARAVANAN

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Lead Data Engineer · AI & Data Platforms · Databricks Certified Professional · Azure · AWS · Kafka · GenAI/RAG

SUMMARY

Lead-level data engineer with 7+ years designing and owning production data and AI platforms across aviation, banking, fintech, and healthcare. Currently building an AI-powered Revenue Intelligence platform for Etihad Airways' Revenue Management team on Azure OpenAI — turning booking, route-performance, and competitive-market data into automated daily insights. Previously built a self-service data ingestion and aggregation platform for a Citi engagement at Scadea and a GenAI/RAG clinical data platform on Databricks serving cited answers over thousands of FDA drug labels with sub-second latency. Deep expertise in distributed systems design, Spark performance optimisation, real-time CDC (Kafka + Debezium), open table formats (Iceberg, Delta Lake), and Unity Catalog-based governance extended across both data and AI assets — from architecture through production ownership.

TECHNICAL SKILLS

Languages:	Python · SQL · Scala · Shell Scripting
Distributed:	Apache Spark (PySpark, Scala) · Apache Kafka · Redpanda · Debezium CDC · Spark Streaming · Apache Flink
Cloud:	Azure Databricks · Delta Lake · Unity Catalog · ADLS Gen2 · ADF · Synapse · AKS · AWS (S3, Redshift, Athena, EMR)
AI / GenAI:	Azure OpenAI · Databricks Mosaic AI · Vector Search · RAG pipelines · LLM tool-calling · Embeddings · Auto Loader
Databases:	PostgreSQL · MongoDB · Cassandra · ScyllaDB · MySQL · Hive
Warehousing:	Snowflake · Amazon Redshift · Apache Iceberg (multi-engine: Spark write / DuckDB read) · Medallion Architecture (Bronze / Silver / Gold)
Transformation:	dbt (dbt-duckdb) · Dimensional / star-schema modeling
Orchestration:	Apache Airflow · Dagster · Databricks Asset Bundles (DAB) · ADF · Oozie
Data Quality:	Great Expectations · dbt tests
DevOps:	GitHub Actions · Docker · Kubernetes (AKS) · Terraform (IaC) · CI/CD · SonarQube
Governance:	Unity Catalog ABAC (data + AI asset governance) · Column-level PII masking · Row-level security · Data contracts · Lineage · Cost attribution

EXPERIENCE

AI Data Platform Engineer — Etihad Airways

Aug 2026 – Present

Revenue Intelligence Platform · Abu Dhabi, UAE

Azure OpenAI · Python · SQL · Email/API Integration · Revenue Management Systems

- Architecting and building an AI-powered Revenue Intelligence platform for Etihad's Revenue Management & Protection team on Azure OpenAI — turning booking, route-performance, and competitive-market data into automated, decision-ready insights.
- Own the platform end-to-end: data ingestion from booking and competitive-intelligence sources, LLM-based analysis and summarisation logic, and a daily email-delivered AI digest — replacing manual route and competitive analysis for revenue management stakeholders.
- Act as the embedded AI/data engineer for revenue management product teams, scoping and shipping new AI intelligence tools as fresh use cases surface — full ownership from architecture through production delivery.

Data Platform Engineer — Scadea

Nov 2025 – Jul 2026

Enterprise Ingestion & Aggregation Platform (Citi engagement) · Chennai, India (Hybrid)

PySpark · Python · Apache Camel (Java) · Hive (Iceberg & Parquet) · Kafka · YARN · Harness · GitHub · SonarQube

- Designed and built a self-service ingestion platform (PySpark + Apache Camel) that auto-detects and parallel-ingests multi-format files across upstream landing zones via control-file triggers into Hive (Iceberg/Parquet) on YARN — eliminating manual ingestion scheduling for every downstream pipeline.
- Engineered fault-tolerant, auto-retrying job orchestration with real-time completion signalling to Kafka, letting downstream consumers react to pipeline state instantly instead of polling — improving reliability and cutting operational overhead across the platform.
- Built a self-serve aggregation framework letting business users define transformation logic directly in SQL — the platform auto-generates Spark execution configs and runs cross-source aggregations (S3, Oracle, Hive) through plug-in connectors, enabling new data sources to be onboarded with zero engineering code changes; standardised the platform's CI/CD on Harness + GitHub with SonarQube-enforced quality gates.

Lead Data Engineer — Wipro

May 2024 – Oct 2025

Risk Engine Consolidation · Starlink Analytics · Chennai, India

Azure Databricks · PySpark · Kafka · Spark Streaming · Apache Airflow · Snowflake · ADF · Scala · AKS · Docker · GitHub Actions

- Risk Evaluation Engine: consolidated 6 fragmented insurance risk pipelines into a single Databricks-native engine — improving claim validation accuracy by 40% and reducing false positives, saving an estimated \$2M+ annually in wrongful claim payouts. Replaced brittle Oozie workflows with observable Apache Airflow DAGs with full retry, alerting, and SLA tracking.
- Starlink Analytics Platform: engineered a Debezium + Kafka → Delta Lake CDC pipeline streaming satellite connectivity change events at sub-60-second end-to-end latency, enabling real-time alerting for 500K+ satellite subscribers. Reduced Spark report generation runtime by 65% via partition pruning, Z-ordering, and Delta caching — unblocking 3 downstream analytics teams from a recurring 4-hour SLA breach.
- Led architecture design reviews and mentored 4 engineers on Spark optimisation, pipeline design patterns, and code quality — improving team PR review cycle by 40%.

Data Engineer — PwC India

Mar 2023 – Feb 2024

Supply Chain Planning & Sales Reporting (PepsiCo engagement) · Bengaluru, India

Azure Synapse · Azure Databricks · ADF · ADLS Gen2 · Azure SQL · PySpark

- Led end-to-end data engineering for a supply chain and sales reporting platform on Azure (ADF + Databricks + ADLS Gen2 + Synapse Analytics), replacing manual reporting workflows with automated, monitored pipelines and near-real-time visibility into inventory and distribution KPIs.
- Owned code reviews, enforced engineering best practices, and drove UAT sign-off ensuring zero production defects at go-live.

Data Engineer — Cartesian

Jan 2022 – Feb 2023

SOLUS.AI — 1:1 Customer Engagement Platform · Bengaluru, India

AWS · PySpark · Python · SQL · Apache Airflow · ADF

- Built large-scale customer behaviour data pipelines on AWS (S3 + PySpark + Redshift) powering SOLUS — a 1:1 personalisation engine processing purchase, browse, and transaction streams to deliver targeted marketing at scale; automated end-to-end ELT workflows with ADF + Airflow, enabling self-serve data delivery for the data science team.

Software Engineer — CIEL

Oct 2018 – Nov 2021

SuprDaily · MediaOcean · IDW (Investment Data Warehouse) · Bengaluru, India

AWS Redshift · Snowflake · Apache Kafka · Apache Airflow · ADF · MongoDB · Athena · PySpark

- Engineered multi-source logistics data pipelines for a last-mile delivery platform on AWS (Redshift, Athena, S3) + Airflow + MongoDB + PySpark, supporting daily operational analytics at scale; separately delivered an Azure media-spend data integration (ADF + ADLS Gen2 + Snowflake + Kafka) with strong SLA adherence across global advertising campaigns.
- Designed and built an Investment Data Warehouse (IDW) — single source of truth for a Swiss bank's wealth management operations, supporting complex financial analytics and audit compliance for 100+ investment analysts managing a \$4B+ AUM portfolio across global markets.

PROJECTS

Clinical Document Intelligence Platform

github.com/saravanaprabhu1398-afk/data-document-intelligence-platform · Databricks · Unity Catalog · Mosaic AI · Vector Search · Auto Loader · Delta Lake · RAG · Python

- Production Databricks Lakehouse converting thousands of unstructured FDA drug labels (PDF/XML) into a queryable AI knowledge base — Auto Loader → Bronze → Silver AI-extracted → Gold dim/fact + Vector Search index — governing the vector index and RAG pipeline under the same Unity Catalog data contracts, lineage, observability, and cost attribution as the underlying data, not as a separate, ungoverned AI layer.
- Serving a cited RAG agent that routes clinical questions to SQL (structured) or Vector Search (unstructured) — returning sub-second answers with source-and-section attribution for auditability.

FlightPulse — Real-Time Flight Delay & Ops Analytics Platform

github.com/saravanaprabhu1398-afk/FlightPulse · Apache Iceberg · dbt · Dagster · Redpanda (Kafka API) · Spark Structured Streaming · Great Expectations · Terraform · MLflow · Streamlit

- Built a local-first, open-source lakehouse platform ingesting live flight positions (OpenSky API) via Kafka-API streaming (Redpanda) and Spark Structured Streaming into Apache Iceberg, plus batch on-time-performance, airport, and weather data via dlt — modeled into a dbt bronze → silver → gold star schema.
- Orchestrated the pipeline as Dagster assets with blocking Great Expectations checks gating every gold-layer table, and proved genuine multi-engine Iceberg access in production — Spark writing the streaming table, DuckDB reading it live via iceberg_scan — with Terraform managing infrastructure and GitHub Actions running CI on every push.
- Trained and MLflow-tracked a flight-delay-probability model (XGBoost vs. logistic-regression baseline) serving a 5-tab Streamlit dashboard against 611K+ real flights and live aircraft data; also root-caused two native production crashes (a DuckDB/pydeck ABI conflict, a concurrent-access SIGBUS).

CERTIFICATIONS

Databricks Certified Data Engineer Professional · Databricks Certified Associate Data Engineer · Microsoft Azure Data Engineer Associate (DP-203) · IBM Certified Data Engineer · Google Cloud Professional Data Engineer (In Progress)

EDUCATION

MBA — Gandhigram Rural Institute, Dindigul, 2018 · B.Sc. Mathematics — Periyar University, Dharmapuri, 2016